

QUBO Formulation: Financial Fraud Feature Selection

Financial Fraud Feature Selection

Problem Statement. Let N denote the number of candidate features. For each feature $i \in \{0, \dots, N-1\}$, let $v_i \in \mathbb{R}$ represent its predictive value. For each pair of distinct features $i, j \in \{0, \dots, N-1\}$ with $i < j$, let $r_{ij} \in \mathbb{R}$ represent the redundancy penalty between them. The objective is to select a subset of exactly K features to maximize the total predictive value minus the total redundancy penalty among selected features. Formally, we seek to maximize

$$O(\mathbf{x}) = \sum_{i=0}^{N-1} v_i x_i - \sum_{0 \leq i < j \leq N-1} r_{ij} x_i x_j \quad (1)$$

subject to the constraint that exactly K features are selected.

Decision Variables. We define binary decision variables $x_i \in \{0, 1\}$ for each candidate feature $i \in \{0, \dots, N-1\}$, where $x_i = 1$ indicates that feature i is included in the selection, and $x_i = 0$ indicates it is excluded.

QUBO Derivation. Step 1: Objective Conversion. The Quadratic Unconstrained Binary Optimization (QUBO) framework minimizes a quadratic Hamiltonian $H(\mathbf{x})$. To convert the maximization problem to minimization, we negate the objective function:

$$H_{\text{obj}}(\mathbf{x}) = -O(\mathbf{x}) = -\sum_{i=0}^{N-1} v_i x_i + \sum_{0 \leq i < j \leq N-1} r_{ij} x_i x_j. \quad (2)$$

Step 2: Constraint Formulation. The requirement to select exactly K features is expressed as the equality constraint:

$$\sum_{i=0}^{N-1} x_i = K. \quad (3)$$

Step 3: Penalty Term Expansion. We enforce the constraint by adding a quadratic penalty term with weight $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} x_i - K \right)^2. \quad (4)$$

Expanding the square and utilizing the idempotent property $x_i^2 = x_i$ for binary variables:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2 \right) \quad (5)$$

$$= \lambda \left(\sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2 \right) \quad (6)$$

$$= \sum_{i=0}^{N-1} \lambda(1 - 2K)x_i + \sum_{0 \leq i < j \leq N-1} 2\lambda x_i x_j + \lambda K^2. \quad (7)$$

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian is the sum of the objective and penalty terms:

$$H(\mathbf{x}) = H_{\text{obj}}(\mathbf{x}) + P(\mathbf{x}) \quad (8)$$

$$= \sum_{i=0}^{N-1} (-v_i + \lambda(1 - 2K)) x_i + \sum_{0 \leq i < j \leq N-1} (r_{ij} + 2\lambda) x_i x_j + \lambda K^2. \quad (9)$$

Step 5: Q Matrix and Offset. Reading off the coefficients from $H(\mathbf{x})$, the symmetric Q matrix entries and constant offset c are given by:

$$Q_{i,i} = -v_i + \lambda(1 - 2K), \quad \forall i \in \{0, \dots, N-1\}, \quad (10)$$

$$Q_{i,j} = r_{ij} + 2\lambda, \quad \forall 0 \leq i < j \leq N-1, \quad (11)$$

$$Q_{j,i} = Q_{i,j}, \quad (12)$$

$$c = \lambda K^2. \quad (13)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any energy reduction gained from the objective function by violating the constraint is outweighed by the penalty cost. Let $M = \max_{\mathbf{x} \in \{0,1\}^N} \left| \sum_{i=0}^{N-1} v_i x_i - \sum_{0 \leq i < j \leq N-1} r_{ij} x_i x_j \right|$ be the maximum absolute value of the objective function over the entire hypercube. Choosing $\lambda > M$ guarantees that for any infeasible assignment, the penalty term $P(\mathbf{x})$ strictly exceeds the maximum possible deviation of the objective term from its minimum, ensuring that infeasible solutions cannot achieve a lower total energy than feasible ones.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal solution to the original constrained problem. For any feasible assignment $\mathbf{x}_{\text{feas}}$, the penalty term vanishes, yielding $H(\mathbf{x}_{\text{feas}}) = -O(\mathbf{x}_{\text{feas}})$. For any infeasible assignment $\mathbf{x}_{\text{infeas}}$, the penalty term is strictly positive. With $\lambda > M$, the penalty cost $P(\mathbf{x}_{\text{infeas}})$ exceeds the maximum possible improvement in $-O(\mathbf{x})$ achievable by any infeasible solution relative to the best feasible solution. Consequently, $H(\mathbf{x}_{\text{infeas}}) > H(\mathbf{x}_{\text{feas}})$ for all feasible $\mathbf{x}_{\text{feas}}$, implying that the global minimum of $H(\mathbf{x})$ must occur at a feasible assignment that maximizes the original objective function.

Illustrative Example. Consider an instance with $N = 3$ candidate features and $K = 2$ features to be selected. Using the general formulas derived above with penalty weight $\lambda = 1000$, we instantiate the Hamiltonian. The penalty expansion yields diagonal coefficients $\lambda(1 - 2K) = 1000(1 - 4) = -3000$, off-diagonal coefficients $2\lambda = 2000$, and offset $\lambda K^2 = 1000(2^2) = 4000$. The resulting QUBO Hamiltonian is:

$$H(\mathbf{x}) = (-v_0 - 3000)x_0 + (-v_1 - 3000)x_1 + (-v_2 - 3000)x_2 \\ + (r_{01} + 2000)x_0x_1 + (r_{02} + 2000)x_0x_2 + (r_{12} + 2000)x_1x_2 + 4000. \quad (14)$$

The corresponding Q matrix entries are $Q_{0,0} = -v_0 - 3000$, $Q_{0,1} = r_{01} + 2000$, and $c = 4000$, with analogous expressions for indices 1 and 2. This formulation encodes the specific instance parameters $N = 3$, $K = 2$, and $\lambda = 1000$ while retaining the predictive values v_i and redundancy penalties r_{ij} as problem-specific parameters.